

TECHNICAL ARTICLE

Rapid understory plant recovery following forest floor protection on temporary drilling pads

Caren E. Jones^{1,2}, Sascha Bachmann¹, Victor J. Lieffers¹, Simon M. Landhäusser¹

Oil sands exploration fragments the boreal landscape by constructing numerous drilling pads to assess underground petroleum reserves. Reclamation of these pads is challenging and slow, particularly for forest understory species. We investigated the feasibility of accelerated forest understory restoration on these temporary pads by taking advantage of the propagule bank and clonal regeneration strategy of many boreal plants. We covered and protected the forest floor (FF) with subsoil during winter pad construction. This forest floor protection (FFP) method was then compared with the current practice of stripping off, stockpiling, and then replacing the FF during the reclamation phase (rollback, RB) and to conventional clearcut (CC) harvesting. In the first growing season, surface disturbance as well as richness and cover of plant regeneration was evaluated; vegetation assessment was repeated in the fifth growing season. Although there were some slight differences between the communities in the FFP and CC treatments, likely associated with varying levels of residual slash and subsoil material, there were striking differences between the communities in the RB and FFP treatments. In addition, while establishment of deciduous tree species was similar between the FFP and CC treatments, there were very few trees found in the RB treatment. The study indicates that protection and careful uncovering of the FF during temporary drilling pad construction should be a technique of choice for forest reclamation used in the boreal forest. However, as RB will still play a part in the reclamation of these sites, management intervention will be required to achieve reclamation goals.

Key words: boreal forest, disturbance, forest restoration, oil sands, soil salvage, vegetation

Implications for Practice

- Protecting the forest floor of boreal mixedwood stands with a subsoil cover during winter construction of temporary drilling pads minimizes damage to propagules of understory species contained in the surface soil.
- Trained machine operators are necessary to construct and deconstruct the forest floor protection treatment. Costs of this treatment should be very similar to current operational procedures of stripping and rolling back forest floor material over an entire pad.
- Where forest soils were protected, woody species and understory plant regeneration and richness was equivalent to areas that experienced forest harvesting typical for the region.
- On the portion of the pad where the traditional rollback was used for reclamation there will likely be a need to artificially introduce woody species.

Introduction

The boreal forest is a disturbance-driven ecosystem where many plants can regenerate vegetatively from sprouting of rhizomes and roots as an adaption to wildfire (Greene et al. 1999; Roberts 2004; Hart & Chen 2006). Increased industrial activities such as resource extraction in this region have introduced new types of disturbances that create challenges for the reestablishment

of forest vegetation communities (Macdonald et al. 2012). Oil sands exploration (OSE) requires the creation of many leveled temporary drilling pads where the forest cover and surface soils need to be removed prior to exploring the oil resource below. After forest removal, the usual construction of an OSE pad entails stripping off the forest floor material (FF—a mixture of the organic forest floor layers and the upper surface mineral soil layer [A horizons]) and stockpiling it off-site. After drilling is completed, the site is recontoured and the salvaged FF is placed back onto the site (rollback [RB]) (Alberta Environment 2005).

Reestablishing both a tree canopy and a viable forest understory community are an essential part of forest reclamation as these components contribute significantly to a range of ecosystem functions (Nilsson & Wardle 2005; Gilliam 2007; Dominati et al. 2010). The recovery of understory plant cover and species richness in forest reclamation sites using RB techniques has

Author contributions: VL, SL, SB conceived, designed, and implemented the experiment; CJ and SB collected and analyzed data; CJ, VL, SL wrote and edited the manuscript; SB, VL, SL wrote and edited an early version of the manuscript.

¹Department of Renewable Resources, University of Alberta, 4-42 Earth Sciences Building, Edmonton, AB T6G 2E3, Canada

²Address correspondence to C. E. Jones, email caren1@ualberta.ca

© 2017 Society for Ecological Restoration

doi: 10.1111/rec.12546

Supporting information at:

<http://onlinelibrary.wiley.com/doi/10.1111/rec.12546/supinfo>

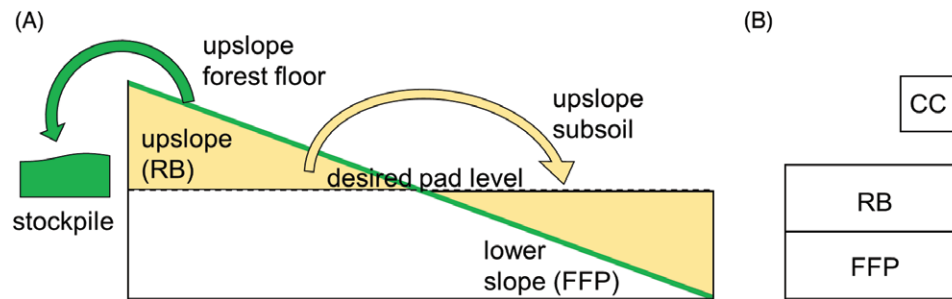


Figure 1. Schematic (A) showing the removal and stockpile of forest floor material and the subsequent relocation of subsoil material from the cut side (rollback [RB]) to the fill side (forest floor protection [FFP]) as well as an aerial view diagram (B) of an experimental block containing a plot of each treatment (RB, FFP, and clearcut [CC]). Adapted from Bachmann et al. 2015.

shown to be challenging, as soil handling damages the vegetative reproductive structures found in the surface soil layers (Tacey & Glossop 1980; Rokich et al. 2000; Mackenzie & Naeth 2010; Macdonald et al. 2015). Furthermore, small root fragments are less likely to sprout than intact roots (Landhäuser & Loeffers 1997; Renkema et al. 2009).

Rather than using the conventional RB reclamation method, a potential viable alternative could be the protection of all or part of the forest floor protection (FFP) during the construction and drilling of the OSE pads, as had been suggested by Osko and Glasgow (2010). This practice would leave below-ground propagules undisturbed and potentially lead to faster restoration of the large number of sprouting species typical of boreal forests. This method has never been rigorously tested in comparison to RB. Furthermore, as most sprouting propagules are left undisturbed in FFP, we expect that its vegetation recovery will be similar to conventional clearcut (CC) forest harvesting, which is commonly practiced in the boreal forest.

The objective of this study was to evaluate plant community redevelopment over five growing seasons after FFP as a mean of improved forest understory vegetation recovery after OSE pad construction and comparing it to conventional RB and CC harvesting in the boreal forest region. We anticipated the RB treatment to have more weedy species colonizing from seed compared to the FFP treatment, which was expected to be dominated by sprouting species typical of the boreal forest; vegetation of the FFP was expected to be similar to sites with a recent aboveground disturbance such as CC harvesting.

Methods

Study Site

The research site is located southeast of Conklin, Alberta (55°24'N, 110°44'W) and in the central mixedwood subregion of the boreal forest region (Natural Regions Committee 2006). This expansive subregion experiences short, cool summers, and long, cold winters, with the majority of precipitation falling during the growing season from May to September (Natural Regions Committee 2006). Average daily temperature for the

region is 2.1°C and average annual precipitation is 421 mm (1987–2010 climate normals; Government of Canada 2017). Forests in this region are fire disturbance driven and are a mosaic of forest stands with varying stand ages and species mixtures dominated by aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* [Moench] Voss). Our research site was dominated by 80-year-old aspen with some white spruce and had understory vegetation typical of a mesic boreal mixedwood dominated by low-bush cranberry (*Viburnum edule* [Michx.] Raf.) (Beckingham & Archibald 1996). The soils are fine-textured, silty sandy Luvisols with clay layers at depth overlaying a morainal till.

A large area (40 ha) was CC harvested commercially in late November of 2011 and six experimental drill pads were installed in late January 2012; clearcutting the pad is a normal preparatory procedure for pad development. Each pad (40 × 60 m) was slightly sloped facing south-southeast.

Treatments

Each of the six pads was split into an upper and lower slope section, each corresponding to a 20 × 60 m plot area. The upper slope portion of the pad received the RB treatment; here the FF was salvaged to 15 cm and stockpiled nearby. The lower slope portion of the pad received the FFP treatment, which was not stripped. The exposed upper slope subsoil in the RB treatment was used to cover the FFP treatment area, creating a leveled area (Fig. 1A). In the FFP, there was no special treatment to delineate FF from the subsoil fill; however, at the time of construction the area had a snow cover (approximately 30 cm). For each pad, a third plot in the CC area was selected adjacent to the RB and FFP plots of the pad (Fig. 1B). This group of three plots was considered the experimental block (replicated six times). To mimic drilling activity, the entire pad was trafficked using a loaded 30-ton rock truck after installation. After lying idle for approximately 3 weeks, the leveled pads were deconstructed and the original FF was uncovered beneath the subsoil. Three weeks after preparation and drilling is a common leave-time before pad deconstruction. The upper slope (RB) was recontoured with the subsoil removed from the lower slope (FFP). The salvaged and stored FF was rough dumped back onto the recontoured upper slope at a similar original depth to salvage and care was taken

Table 1. Average vegetation cover and richness in the clearcut (CC), forest floor protection (FFP), and rollback (RB) treatments in 2012 and 2016. Data is further organized into the proportion of the total cover and species richness by species of various functional groups (forbs, graminoids, and shrubs). Means ($\pm$ SE) with different letters for the treatments indicate statistically significant differences ($\alpha = 0.1$) across the table ($n = 6$). Standard errors are not shown for the proportion values as they were all ≤ 0.1 . * indicates data where the Box-Cox transformation was applied prior to analysis.

	CC		FFP		RB	
	2012	2016	2012	2016	2012	2016
Vegetation cover*	29.5 _B (3.5)	53.4 _A (4.7)	30.9 _B (3.1)	65.0 _A (6.4)	4.6 _C (0.5)	56.9 _A (5.2)
Species richness	10.4 _B (0.8)	10.8 _{AB} (0.5)	10.7 _B (0.9)	13.0 _A (0.5)	6.3 _C (0.3)	10.6 _B (0.4)
<i>Proportion of cover</i>						
Forbs*	0.57 _B	0.57 _B	0.57 _B	0.57 _B	0.85 _A	0.63 _B
Graminoids*	0.18 _A	0.12 _{ABC}	0.15 _{AB}	0.08 _{BC}	0.05 _C	0.09 _{ABC}
Shrubs*	0.24 _A	0.30 _A	0.27 _A	0.34 _A	0.10 _B	0.27 _A
<i>Proportion of richness</i>						
Forbs*	0.66 _B	0.61 _B	0.64 _B	0.59 _C	0.81 _A	0.64 _B
Graminoids*	0.09 _{BC}	0.10 _{AB}	0.09 _{BC}	0.09 _{AB}	0.05 _C	0.13 _A
Shrubs*	0.21 _A	0.23 _A	0.23 _A	0.26 _A	0.13 _B	0.18 _B

to minimize machine traffic during RB of the FF. Experimental pad construction and deconstruction were complete in March 2012. Further details regarding site construction can be found in Bachmann et al. (2015).

Measurements

Three transects were installed perpendicular to the slope in each of the three treatment plots; each transect had four sample points spaced 4 m apart. Measurements of site conditions were done in each of the 12 subplots in August 2012 and included measures of woody debris and subsoil residue cover as well as evaluations (yes/no) of surface soil disturbance (Table S2, Supporting Information).

Vegetation assessments were also made in these subplots in the three treatments. Sampling was carried out in late August of 2012 and again in late July 2016. Plant cover was estimated by species in a 1 m² quadrat using seven cover classes: <1%, 1–3, 3–5, 5–10, 10–25, 25–50, 50–75, and >75% (cover); all plants were identified to species level (Table S5). Nomenclature and associated authorities follow Moss (1994). Density and maximum height measurements of deciduous tree regeneration from root suckers were also made in these subplots in both 2012 and 2016.

Data Analyses

The estimated plant covers within each subplot were converted to midpoint values prior to computing averages (Macdonald & Fenniak 2007). Plant species were also categorized into growth form functional groups (tree, shrub, forb, graminoid). Species richness per subplot was determined by converting the cover data into binary data and then calculating number of species found in each subplot; richness was then also determined per functional group for each subplot.

Total plant cover, total species richness, and proportion of cover and richness by functional group were averaged for each treatment plot prior to analyses. Data that did not meet normality and homogeneity of variance requirements were Box-Cox

transformed (Table 1). Linear mixed effect models with repeated measures were run for each response category ($n = 6$) with year, treatment, and a year by treatment interaction being tested using R software (version 3.2.2; R Core Team 2017). When there was a significant effect or interaction ($\alpha = 0.1$; selected based on small sample size), the analysis was followed up by pairwise comparisons with Tukey adjusted alphas.

The effects of the three treatments on the 2012 and 2016 plant communities were analyzed using a permutational multivariate analysis of variance (PERMANOVA). When statistically significant effects were found ($\alpha = 0.1$), post hoc tests were done with Holm adjusted alphas. Indicator species analyses were used to determine which species drove the significant differences identified in the PERMANOVA. Single species cover was also used in nonmetric multidimensional scaling (NMDS) analysis to produce an ordination of plant community structure among the three treatments in 2012 and 2016. The NMDS was executed on an untransformed data matrix of species cover with a random starting configuration, a stability criterion of 0.00005, the Bray distance measure, and the Wisconsin-style double standardized scaling. Two dimensions were used for the final graph of the NMDS, including ellipses showing 90% confidence intervals around the centroid of each group. In addition, correlation of the main treatment ordination with deciduous tree cover was also calculated and plotted on the NMDS. To further explore the variation in the FFP and RB in 2016, average single species cover per block were used to run an additional NMDS.

Linear mixed effect model analysis was used to assess the impact of cover site factors (slash [FFP and CC] and residual subsoil [FFP only]) on total plant cover and species richness in 2016. For significant interactions or effects ($\alpha = 0.1$), it was then tested if all functional groups were also significantly influenced. Linear mixed effect models were also used to determine the effect of deciduous tree cover on total plant cover and species richness in 2012 and 2016 in the FFP and CC treatments. When significant interactions or effects were found ($\alpha = 0.1$), the response of functional groups was subsequently tested. The effect of surface soil disturbance on plant cover and richness



Figure 2. Vegetation recovery during the first (2012) and fifth (2016) growing seasons. The left side of each image shows the rollback treatment whereas the right of each image shows the forest floor protection treatment.

(total and per functional group) was assessed with two tailed *t*-tests.

Results

Impact of Treatments

At the end of the first growing season in 2012, there were striking differences in the visual development of vegetation cover between the RB and FFP treatments, and whereas these differences in cover were less evident in 2016 (Fig. 2), the plant community developing in the RB treatment was still significantly different from the communities developing in the CC ($p < 0.005$) or FFP ($p < 0.005$) treatments (Fig. 3). While total vegetation cover and species richness was significantly lower in the RB treatment than the CC ($p < 0.001$) and FFP ($p < 0.001$) treatments in 2012, by 2016 cover and richness were similar among all three treatments (Table 1). Deciduous tree cover did not significantly change in the RB treatment between 2012 and 2016 and was significantly lower than in the CC ($p < 0.001$ both years) or FFP treatments ($p < 0.001$ both years). Density of deciduous trees significantly decreased (all $p < 0.1$) whereas maximum tree height increased (all $p < 0.001$) from 2012 to 2016 in all three treatments (Table S1).

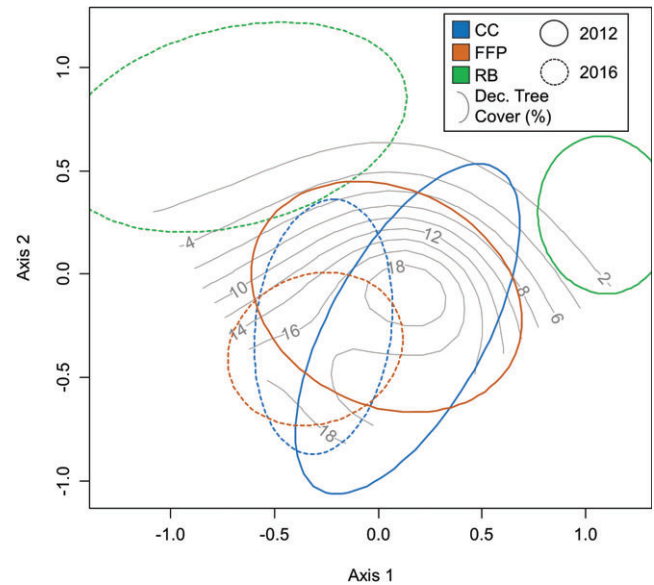


Figure 3. Nonmetric multidimensional scaling (NMDS) ordination of the vegetation community composition in the clearcut (CC), forest floor protection (FFP), and rollback (RB) treatments in the first (2012) and fifth (2016) growing seasons (stress = 0.16). Ellipses delineate a 90% confidence interval around the centroid of the different treatments for the different years (blue, CC; orange, FFP; green, RB; solid line, 2012; dashed line, 2016). Gray background lines and values indicate cover (%) by deciduous trees.

FFP and CC Treatments

Total cover, total richness, and proportion of cover by forbs and shrubs were not significantly different between the FFP and CC treatments in either year measured (Table 1), and the communities in both the CC and FFP treatments in 2012 and 2016 were associated with high deciduous tree cover (Fig. 3). Although similar in 2012, proportion of richness by forbs was significantly lower in 2016 in the FFP treatment than in the CC treatment ($p < 0.05$; Table 1). The overall communities were also significantly different in both 2012 and 2016 between the two treatments (both years $p < 0.05$; Fig. 3). While neither of the treatments had significant indicator species in their 2012 communities, in 2016 the CC treatment had two indicator species (*Maianthemum canadense* Desf. and *Linnaea borealis* L.), which are forest-associated species whereas the FFP had a greater total number of indicator species that also included various forest-associated species indicative of the original forest community (e.g. *Viburnum edule*, *Vaccinium vitis-idaea* L.; Table S3).

The 2016 vegetation communities in the FFP ($p < 0.005$) and CC ($p < 0.005$) treatments were also significantly different than their 2012 predecessor. From 2012 to 2016, the communities within the CC and FFP treatments became more homogenous as shown by the reduction in the size of the 90% confidence intervals associated with each treatment in the NMDS (Fig. 3). Although the communities in the two growing seasons were statistically different for the two treatments, they occupy similar areas of the NMDS (Fig. 3), suggesting that while there are

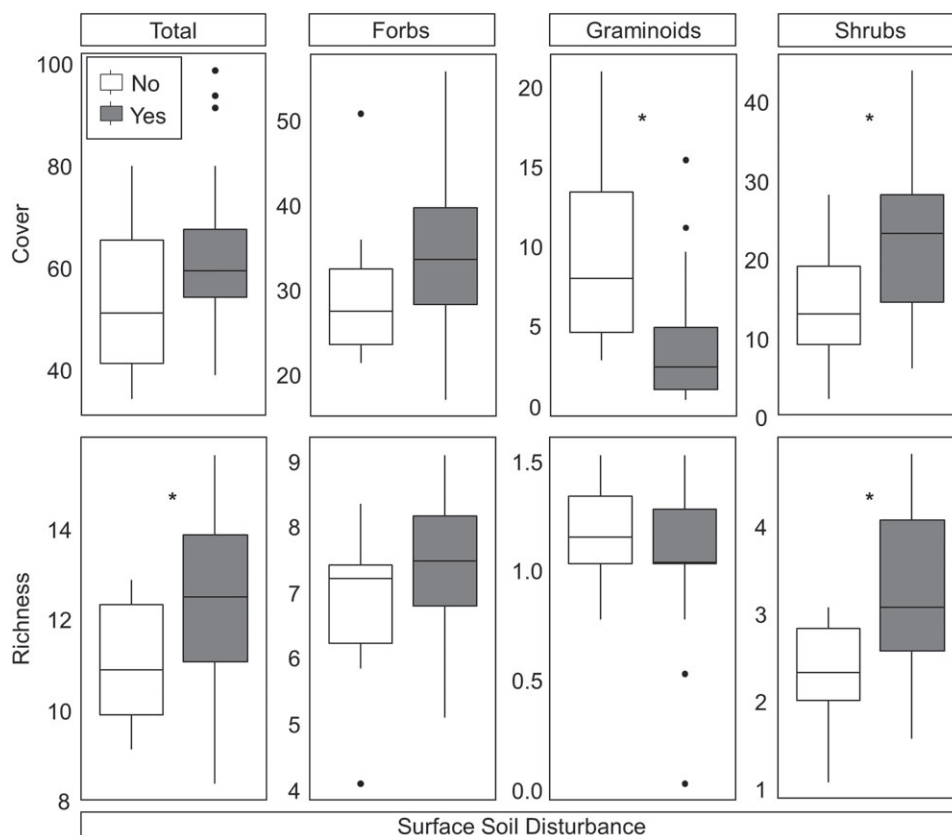


Figure 4. Average vegetation cover and richness in the 2016 clearcut and forest floor protection treatments combined, in relation to surface soil disturbance. Cover and richness are shown as totals (all vegetation) and then split into forb species, graminoid species, and shrub species. A significant effect of surface soil disturbance is indicated with an asterisk ($\alpha = 0.1$).

species which are more dominant in each year (Table S3), there are many similarities in terms of overall species composition between them.

On a subplot basis in both the CC and FFP, there was a significant interaction between year and deciduous tree cover (primarily *Populus tremuloides* but also included *Populus balsamifera* L. and *Betula papyrifera* Marshall) on total vegetation cover ($p < 0.005$) and total species richness ($p < 0.05$). Tree cover was significantly associated with higher vegetation cover in 2016 ($R^2 = 0.09$, $p < 0.05$) and reduced species richness in 2012 ($R^2 = 0.07$, $p < 0.1$). Increased tree cover was significantly associated with reduced graminoid cover ($R^2 = 0.16$, $p < 0.001$), reduced shrub richness in 2012 ($R^2 = 0.06$, $p < 0.1$), and reduced shrub cover in both 2012 ($R^2 = 0.09$, $p < 0.05$) and 2016 ($R^2 = 0.12$, $p < 0.05$; Table S4).

Subplots with surface soil disturbance had significantly greater shrub cover ($p < 0.05$), overall richness ($p < 0.1$), particularly of shrub species ($p < 0.05$), and significantly lower graminoid cover ($p < 0.05$) in 2016 (Fig. 4). Subplots in the FFP and CC treatments with higher slash cover were associated with lower vegetation cover ($R^2 = 0.17$, $p < 0.005$). Note that we combined the FFP and CC treatments for these analyses as there was no significant interaction between treatment and surface soil

disturbance/slash cover on the vegetation in 2016, allowing an increase in the power of the analysis through the increased sample size. Subplots in the FFP treatment had significantly lower total species richness ($R^2 = 0.25$, $p < 0.05$) in 2016, particularly by forb species ($R^2 = 0.20$, $p < 0.1$), when residual subsoil cover was greater.

Rollback Treatment

Between 2012 and 2016, the community in the RB treatment also changed significantly ($p < 0.005$). Total cover ($p < 0.001$), species richness ($p < 0.001$), and proportion of cover by shrubs ($p < 0.005$) significantly increased between measurement years, whereas proportion of cover ($p < 0.005$) and richness ($p < 0.001$) by forbs significantly decreased (Table 1). The 2012 community was characterized by a native, herbaceous, annual species (*Vicia americana* Muhl. ex Wild.) whereas the 2016 community had several native forb and shrub indicator species (e.g. *Epilobium angustifolium* (L.) Holub, *Rubus idaeus* L.) as well as introduced forb species (e.g. *Crepis tectorum* L., *Taraxicum officinale* F. H. Wigg., *Trifolium hybridum* L.; Table S3). Unlike the communities in the FFP treatment, the vegetation community in the RB became more heterogeneous between 2012 and 2016 ($p < 0.005$; Fig. 3). When the 2016 RB community was further explored, this

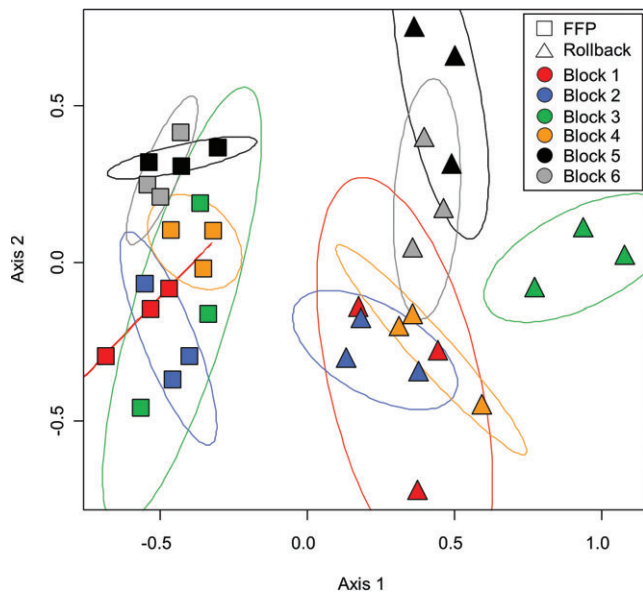


Figure 5. Nonmetric multidimensional scaling (NMDS) ordination depicting the vegetation community composition in the rollback (RB; triangles) and forest floor protection (FFP; squares) treatments in the fifth growing season (stress = 0.15). Ellipses indicate vegetation communities (90% confidence intervals) in the six blocks (colors) in the two treatments.

increased heterogeneity was a result of the communities in two of the blocks (blocks 3 and 5) diverging from the other blocks (Fig. 5).

Discussion

Protecting, rather than stripping the FF material during OSE pad construction resulted in greater forest understory species and aspen recovery. This different approach in the construction and reclamation of OSE pads resulted in vegetation communities similar to communities recovering from aboveground disturbance such as CC harvesting, a disturbance better understood in boreal forest recovery. CC and FFP treatments had relatively minimal disruption and handling of the FF, leaving the soil propagule bank largely undisturbed, allowing species adapted to aboveground disturbance to recover quickly. The RB treatment, however, produced severe soil disturbance and likely diluted, damaged, or buried propagules during salvage and placement (Rokich et al. 2000; Holmes 2001; Wachowski et al. 2014), resulting in little to no aspen regeneration and few native forest understory species. Dense beds of *Rubus idaeus* and *Astragalus canadensis* L. will likely continue to reduce establishment and development of forest understory and tree species in the treatment.

Similarly to the CC harvested sites, FFP had rapid initial recovery of aspen (Bachmann et al. 2015), which provided shade and sheltered conditions for obligate understory species (e.g. *Cornus canadensis* L., *Mitella nuda* L.) and species commonly found in mature aspen forests (e.g. *Viburnum edule*, *Aralia nudicaulis* L.) emerging from the undisturbed

soil propagule bank (Liefers 1995; Macdonald & Fenniak 2007). Areas with greater deciduous cover were also associated with higher overall vegetation cover and richness, particularly by shrub species; there was also lower graminoid cover in areas with high deciduous cover. Reestablishment of shrubs is important for the success of the proposed reclamation method as they are significant drivers of understory vegetation development in newly disturbed habitats by establishing an early canopy (Peterson & Reich 2001; Sorenson et al. 2011). Graminoid species can create unfavorable conditions during boreal forest regeneration through increased belowground competition (Kozłowski 2002; Balandier et al. 2005), therefore minimizing their establishment is often desirable.

Not only did the deciduous tree canopy influence the type of species that developed in the understory community, it may also have mitigated the influence and impact of site conditions. Divergence in the distribution of the confidence interval ellipses for the RB blocks in the NMDS, particularly blocks 3 and 5, suggests greater community variability between the blocks in the RB treatment than the FFP blocks based on their comparatively tight ellipses distribution. Tree canopies provide shelter that controls light, temperature, and humidity regimes in the understory, which all are linked to various edaphic conditions in the understory (Barbier et al. 2008), and while it is unclear what site factors specifically were causing the RB communities to diverge between the blocks, the presence of a closed canopy in the adjacent FFP plots appears to have homogenized site conditions.

While the overall structure of vegetation communities in the FFP and CC treatments was similar, the two communities had some floristic differences, likely a result of the varying amounts of woody debris and residual subsoil in the treatments. While woody debris has been found to create microsites beneficial for vegetation germination and development on bare reclamation sites (Brown & Naeth 2014), increased woody debris cover in the FFP treatment was found to decrease aspen sucker density in the first growing season (Bachmann et al. 2015) and decreased vegetation cover in the fifth growing season. Soil temperature was cooler under the slash cover (Bachmann et al. 2015), potentially slowing vegetation development (DesRochers et al. 2002; Landhäusser et al. 2007). Residual subsoil material in the FFP may also have contributed to the differences between the communities in the treatments. In areas where operators were less successful at removing the entire subsoil cap, species richness, particularly by herbaceous species, was lower. Although the depth of the residual subsoil was not measured in our study, the negative impact of the residual material could be the result of excessive burial of propagules as this has been found to reduce germination and emergence (Putwain & Gillham 1990; Macdonald et al. 2015).

To reduce residual subsoil material covering the FF in the FFP treatment, operators might cause more disturbance to the protected FF layer. While this may damage viable propagules in the surface soil layer, we found that limited surface soil disturbance positively impacted shrub cover, total species richness, and shrub richness while reducing graminoid cover. Heterogeneous soil surface conditions have shown to result in a larger

range of microsites that can facilitate the establishment of a greater array of species (Johnson & Fryer 1992; Biederman & Whisenant 2011). Therefore, although there is potentially a trade-off between leaving or removing all subsoil capping material at the risk of damaging the protected FF, some disturbance to the FF layers appears to not adversely affect vegetation recovery.

The FFP technique resulted in rapid vegetation recovery of forest understories on the OSE pad, but the commonly used RB technique with its stripping, salvaging, and replacement of FF resulted in very delayed recovery. The RB technique may lead to relatively stable communities dominated by nonforest associated shrubs and/or grasses that will be slow to develop into a forest. As such, tree planting may be needed in these areas to achieve desired forest reclamation goals. In addition, the application of the FFP technique was largely successful because most species in these forests can rapidly sprout from roots and rhizomes, and as such, application of this technique may not be appropriate in forest types where the majority of species do not regenerate vegetatively.

Acknowledgments

We would like to thank G. Sturt, T. Saad, C. Kilba, C. Walsh, D. Pope, and M. Dyck who assisted in the project. We are especially grateful to E. Marenholz, F. Leishman, and R. Caners for helping to set up the project, as well as to A. Snively, P. Lefebvre, L. Dent, S. Stack, and D. Cirellii for field work assistance. Devon Energy provided direct and in-kind support and the National Science and Engineering Research Council (NSERC) through a CRD grant.

LITERATURE CITED

- Alberta Environment (2005) Code of Practice for Exploration Operations (made under the Environmental Protection and Enhancement Act, RSA 2000, c.E-12, as amended and Conservation and Reclamation Regulation (AR 115/93), as amended). Alberta Queen's Printer, Edmonton, Canada
- Bachmann S, Lieffers VJ, Landhäusser SM (2015) Forest floor protection during drilling pad construction promotes resprouting of aspen. *Ecological Engineering* 75:9–15
- Balandier P, Collet C, Miller JH, Reynolds PE, Zedaker SM (2005) Designing forest vegetation management strategies based on the mechanisms and dynamics of crop tree competition by neighbouring vegetation. *Forestry* 79:3–27
- Barbier S, Gosselin F, Balandier P (2008) Influence of tree species on understory vegetation diversity and mechanisms involved—a critical review for temperate and boreal forests. *Forest Ecology and Management* 254:1–15
- Beckingham JD, Archibald JH (1996) Field guide to ecosites of Northern Alberta. Special Report No. 5, Natural Resources Canada, Canadian Forest Service, Northern Forestry Centre, Edmonton, Canada
- Biederman LA, Whisenant SG (2011) Using mounds to create microtopography alters plant community development early in restoration. *Restoration Ecology* 19:53–61
- Brown RL, Naeth MA (2014) Woody debris amendment enhances reclamation after oil sands mining in Alberta, Canada. *Restoration Ecology* 22:40–48
- DesRochers A, Landhäusser SM, Lieffers VJ (2002) Coarse and fine root respiration in aspen (*Populus tremuloides*). *Tree Physiology* 22:725–732
- Dominati E, Patterson M, Mackay A (2010) A framework for classifying and quantifying the natural capital and ecosystem services of soils. *Ecological Economics* 69:1858–1868
- Gilliam FS (2007) The ecological significance of the herbaceous layer in temperate forest ecosystems. *BioScience* 57:845–858
- Government of Canada (2017) Canadian climate normals. http://climate.weather.gc.ca/climate_normals/index_e.html (accessed 3 Apr 2017)
- Greene DF, Zassada JC, Sirois L, Kneeshaw D, Morin H, Charron I, Simard M-J (1999) A review of the regeneration dynamics of North American boreal forest tree species. *Canadian Journal of Botany* 29:824–839
- Hart SA, Chen HYH (2006) Understory vegetation dynamics of North American boreal forests. *Critical Reviews in Plant Sciences* 25:381–397
- Holmes PM (2001) Shrubland restoration following woody alien invasion and mining: effects of topsoil depth, seed source, and fertilizer addition. *Restoration Ecology* 9:71–84
- Johnson EA, Fryer GI (1992) Physical characterization of seed microsites – movement on the ground. *Journal of Ecology* 80:823–836
- Kozłowski TT (2002) Physiological ecology of natural regeneration of harvested and disturbed forest stands: implications for forest management. *Forest Ecology and Management* 158:195–221
- Landhäusser SM, Lieffers VJ (1997) Seasonal changes in carbohydrate storage and regrowth in rhizomes and stems of four boreal forest shrubs: applications in *Picea glauca* understory regeneration. *Scandinavian Journal of Forest Research* 12:27–32
- Landhäusser SM, Lieffers VJ, Chow P (2007) Impact of chipping residues and their leachate on the initiation and growth of aspen root suckers. *Canadian Journal of Soil Science* 87:361–367
- Lieffers VJ (1995) Ecology and dynamics of boreal understory species and their role in partial-cut silviculture. Pages 33–39. In: Bamsey CR (ed), Innovative silviculture systems in boreal forests. Clear Lake Ltd., Edmonton, Canada
- Macdonald SE, Fenniak TE (2007) Understory plant communities of boreal mixedwood forests in western Canada: natural patterns and response to variable-retention harvesting. *Forest Ecology and Management* 242:34–48
- Macdonald SE, Quideau S, Landhäusser SM (2012) Rebuilding boreal forest ecosystems after industrial disturbance. Pages 123–160. In: Vitt DH, Bhatti JS (eds) Restoration and reclamation of boreal ecosystems: attaining sustainable development. Cambridge University Press, Edmonton, Canada
- Macdonald SE, Snively AK, Fair JM, Landhäusser SM (2015) Early trajectories of forest understory development on reclamation sites: influence of forest floor placement and a cover crop. *Restoration Ecology* 23: 698–706
- Mackenzie DD, Naeth MA (2010) The role of the forest soil propagule bank in assisted natural recovery after oil sands mining. *Restoration Ecology* 18:418–427
- Moss EH (1994) In: Packer JG (ed) The flora of Alberta. 2nd edition. University of Toronto Press, Toronto, Canada
- Natural Regions Committee (2006) Downing DJ, Pettapiece WW (eds) Pub. No. T/852 Natural regions and subregions of Alberta. Government of Alberta
- Nilsson MC, Wardle DA (2005) Understory vegetation as a forest ecosystem driver: evidence from the northern Swedish boreal forest. *Frontiers in Ecology and the Environment* 3:421–428
- Osko T, Glasgow M (2010) Removing the wellsite footprint: recommended practices for construction and reclamation of wellsites on Upland Forests in Boreal Alberta, Edmonton, Canada Department of Biological Sciences University of Alberta
- Peterson DW, Reich PB (2001) Prescribed fire in Oak Savanna: fire frequency effects on stand structure and dynamics. *Ecological Applications* 11:914–927
- Putwain PD, Gillham DA (1990) The significance of the dormant viable seed bank in the restoration of heathlands. *Biological Conservation* 52:1–16
- R Core Team (2017) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <http://www.R-project.org/>
- Renkema KN, Landhäusser SM, Lieffers VJ (2009) Suckering response of aspen to traffic-induced-root wounding and the barrier-effect of log storage. *Forest Ecology and Management* 258:2083–2089

- Roberts MR (2004) Response of the herbaceous layer to natural disturbance in North American forests. *Canadian Journal of Botany* 82: 1273–1283
- Rokich DP, Dixon KW, Sivasithamparam K, Meney KA (2000) Topsoil handling and storage effects on woodland restoration in Western Australia. *Restoration Ecology* 8:196–208
- Sorenson PT, Quideau SA, MacKenzie MD, Landhäusser SM, Oh SW (2011) Forest floor development and biochemical properties in reconstructed boreal forest soils. *Applied Soil Ecology* 49:139–147
- Tacey WH, Glossop BL (1980) Assessment of topsoil handling techniques for rehabilitation of sites mined for bauxite within the Jarrah Forest of Western Australia. *Journal of Applied Ecology* 17:195–201
- Wachowski J, Landhäusser SM, Lieffers VJ (2014) Depth of root placement, root size and carbon reserves determine reproduction success of aspen root fragments. *Forest Ecology and Management* 313:83–90

Coordinating Editor: Andrew Grigg

Supporting Information

The following information may be found in the online version of this article:

Table S1. Average deciduous tree density, cover, and maximum height ($\pm$ SE) in the clearcut (CC), forest floor protection (FFP), and rollback (RB) treatments in 2012 and 2016 ($n = 6$).

Table S2. Edaphic factors per treatment in 2012.

Table S3. Results of indicator species analysis on vegetation composition data (percent cover by species) in the first and fifth growing season for the three.

Table S4. Linear mixed effect regression analysis results for plant cover and species richness in the clearcut and forest floor protection treatments in 2012 and 2016 in relation to deciduous tree cover.

Table S5. Complete list of species found in the vegetation surveys in 2012 and 2016. Included is which treatment the species was found in (clearcut [CC], rollback [RB], forest floor protection [FFP]) and its growth form functional group (graminoid [G], forb [F], shrub [S], tree [T], and nonvascular [NV]).

Received: 27 February, 2017; First decision: 1 April, 2017; Revised: 25 April, 2017; Accepted: 25 April, 2017; First published online: 21 August, 2017